

11. Write a Java program to perform JDBC operations (connect, insert, update, delete)

Type	Input	Expected Output	Explanation
Success	Connect to database	Connection successful	JDBC connection
Success	Insert employee record	Record inserted	Insert operation
Success	Update employee salary	Record updated	Update operation
Failure/Edge	Delete non-existing employee ID	0 rows affected	Safe delete
Failure/Edge	Insert duplicate primary key	SQL exception	Constraint validation

Algorithm:

1. **Start**
2. Import the required packages:
 - o `java.sql.*` for JDBC operations.
 - o `java.util.*` for Scanner.
3. Define the database details:
 - o Database URL
 - o Username
 - o Password
4. Load the MySQL JDBC driver using `Class.forName()`.
5. Establish a connection with the MySQL database using `DriverManager.getConnection()`.
6. Create a `Statement` object.
7. Create a `Scanner` object to read the user's choice.
8. Set a Boolean variable `a = true` to start the menu loop.
9. Display the menu:
 - o `i` → Insert
 - o `u` → Update
 - o `d` → Delete
 - o `s` → Select
 - o `e` → Exit
10. Read the user's choice.
11. Perform the selected operation:
 - **If choice is `i`:**
 1. Create an `INSERT` SQL query using `PreparedStatement`.
 2. Set the email, password, username, address, and phone number values.
 3. Execute the query using `executeUpdate()`.
 4. Display an insertion-success message.
 - **If choice is `u`:**
 1. Create an `UPDATE` SQL query.
 2. Change the username where the phone number is 123.
 3. Execute the query.
 - **If choice is `d`:**
 1. Create a `DELETE` SQL query.
 2. Delete the record whose phone number is 123.
 3. Display the number of affected records.
 - **If choice is `s`:**
 1. Execute a `SELECT *` query.
 2. Store the result in a `ResultSet`.
 3. Read each record using `rs.next()`.

4. Display the retrieved values.
 - **If choice is e:**
 1. Set `a = false`.
 2. Exit the loop.
 - **Otherwise:** Display "enter correct choice".
12. Continue displaying the menu until the user selects e.
 13. If an exception occurs, catch it and display the error using `printStackTrace()`.
 14. **Stop.**

Program:

```
import java.sql.*;
import java.util.*;

public class Week11 {

    static final String DB_URL = "jdbc:mysql://localhost:3306/login1";
    static final String USER = "root";
    static final String PASS = "";

    public static void main(String[] args) {

        // Open a connection
        try
        {
            Class.forName("com.mysql.jdbc.Driver");
            Connection conn = DriverManager.getConnection(DB_URL, USER, PASS);
            Statement stmt = conn.createStatement();
            Scanner sc = new Scanner(System.in);
            boolean a = true;
            while(a)
            {
                System.out.println("entr choice");
                System.out.println("i:insert,u:update,d:delete,s:select,e:exit");
                String ch = sc.next();

                switch (ch) {
                    case "i":
                    {

                        // Execute a query
                        System.out.println("Inserting records into the table...");
                        String email = "aaaab", pass = "bbbbc", username = "cccd", add = "eeef";
                        int phno = 123;

                        PreparedStatement ps = conn.prepareStatement("INSERT INTO login VALUES (?, ?, ?, ?, ?)");
                        ps.setString(1, email);
                        ps.setString(2, pass);
                        ps.setString(3, username);
                        ps.setInt(5, phno);
```

```

ps.setString(4, add);
int i=ps.executeUpdate();

        System.out.println("Inserted records into the table...");
        break;
    }
    case "u":
    {
        String sql = "UPDATE login " +
            "SET username = 'yyyyy' WHERE phno in (123)";
        stmt.executeUpdate(sql);
        break;
    }
    case "d":
    {
        int result=stmt.executeUpdate("delete from login where phno=123");
        System.out.println(result+" records affected");
        break;
    }
    case "s":
    {
        ResultSet rs=stmt.executeQuery("select * from login");
        while(rs.next())
        System.out.println(rs.getString(1)+" "+rs.getString(2)+" "+rs.getString(3)+"
            "+rs.getInt(4)+" "+rs.getString(5));
        break;
    }
    case "e":
    {
        a=false;
    }
    default:
    {
        System.out.println("enter correct choice");
    }
    }
} }catch (Exception e) {
    e.printStackTrace();
}

    }
}

```